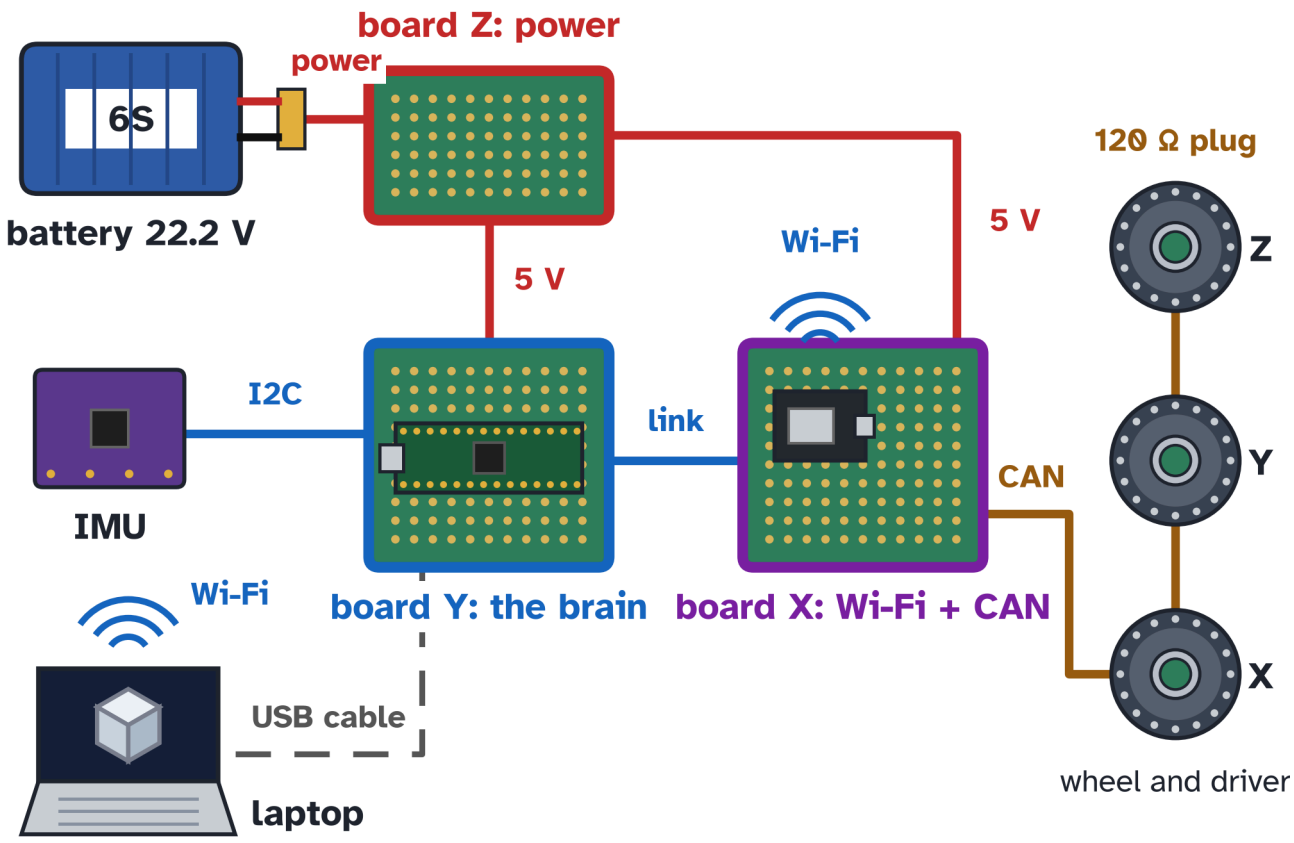
Wishlist

for the next upgrade

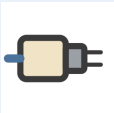
What the build and the tests of October 2026 taught - collected for Cubi 2's next version.



Designed by Athanasia Nikolova

All data come from tests and many real builds, not only from theory. v1.2 · November 2026

A must - for the next upgrade



USB sockets reachable from outside

On the boards the USB sockets face into the cube: plugging in a cable takes a long time. Turn them outwards, reachable with the cube built.



A place for balance weights

A mounting to fit small weights, to set the cube's balance.

The build



A stiffer battery and IMU column

A thicker battery tube (Ø13.5 mm, about 3.7 times stiffer) printed at 100 % infill, or the IMU table of the two concepts (report/concepts). The basin closed on its open side.



A strong box standoff at corner 0

The first one broke at its flange ribs (26/10). Keep the reinforced one and check it after long runs.



CAN plugs reachable with the wheels on

The driver ids are set one driver at a time: today the plugs hide behind the wheels once the cube is built.



More current for the drivers

8 A today; 12 - 15 A lets the sphere climb from an opening onto a navel.

The programs and the app



The accelerometer calibrated on 6 faces

In cubi2_setup: then 'zero hang' gives the true balance point at once.



First set-up over Bluetooth

The Wi-Fi board shows up as 'Cubi2-setup-XXXX' with a PIN: a new cube is set up without a USB cable.



The sphere's openings calibrate the IMU

Each of the 24 openings holds the sphere at a known direction.



'climb N'

From an opening onto navel N, with the motors only (needs the higher current).



The Electron app, complete

Update, the cube's questions, the last run, the Guide and the profiles, as in the Cubi 2 app - then its own new features.



Cubi 2 on spaceinclass.com

The simulation online; the choice of how is still open.

To measure



The 5 V line

y1 - y2 read 5.0 V (4.5 - 4.7 V expected): measure z11 - z10, before the diode, with the battery in.

From the test log and the notes of the build (cubi2.json).